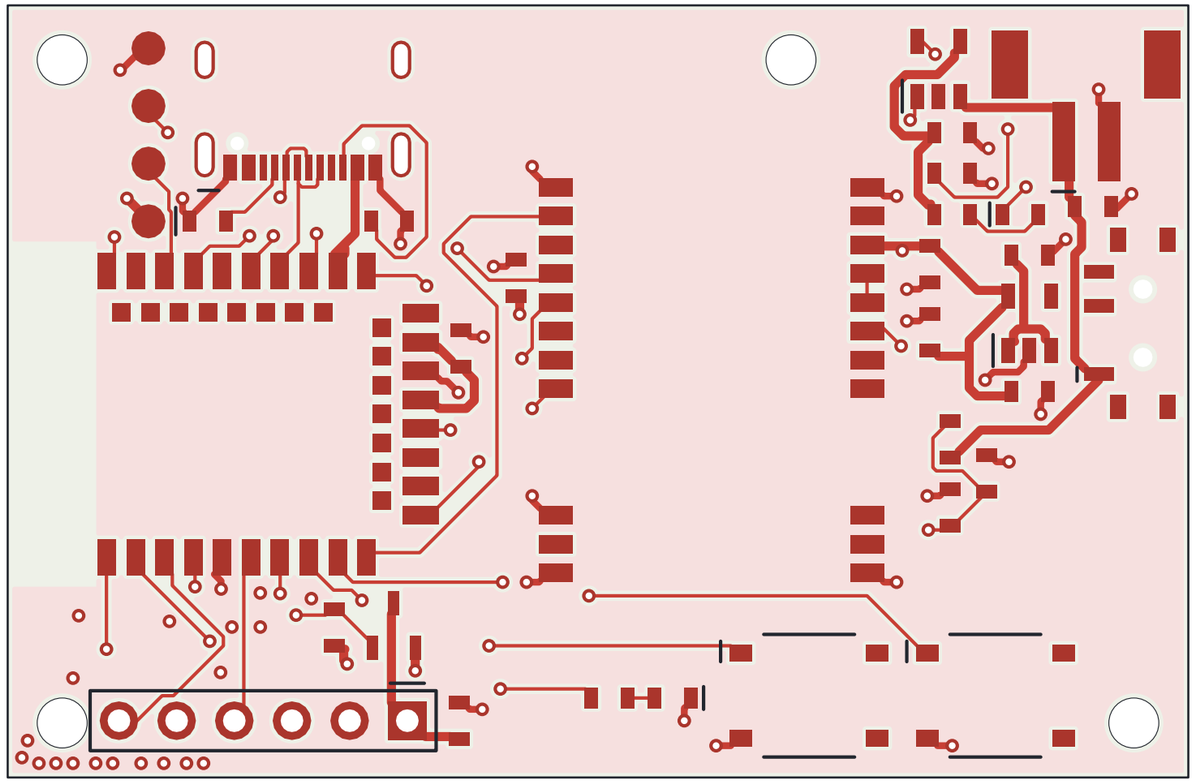
LoRa Mesh Node Mini

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1S LiPo, GPS header

REVISION	DATE	AUTHOR	COMPANY
A	2026-11-02	SolidCraft example	–



Top side · board 52.0 × 34.0 mm, 2 copper layers, 30 parts fitted on both sides

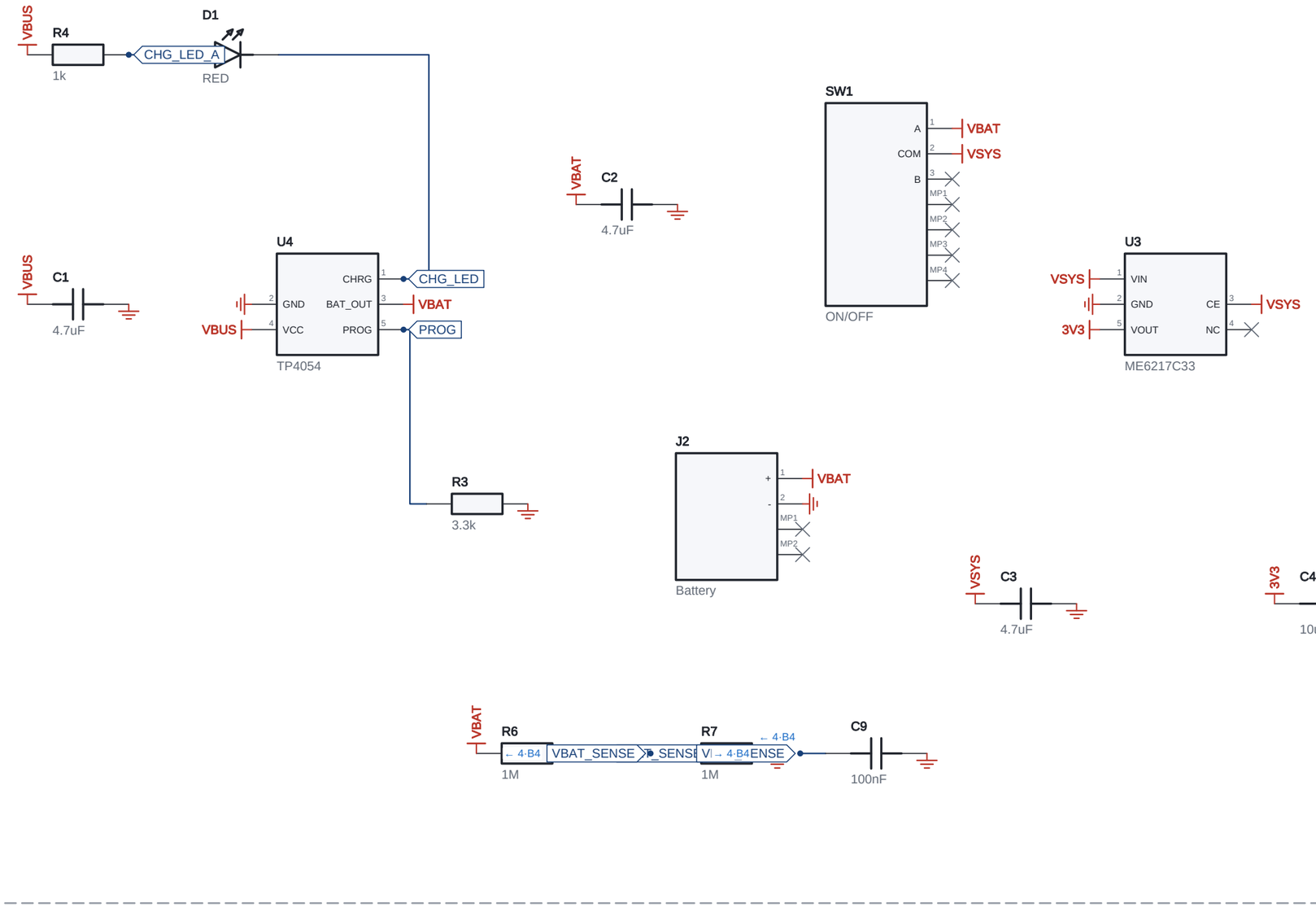
Contents

2 Power
3 USB
4 MCU
5 LoRa
6 GPS
7 UI
8 Block diagram
9 Parts list (1 of 2)
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11 Board top
12 Board bottom (seen from the bottom)
13 Assembly top
14 Assembly bottom (seen from the bottom)
15 Layer stack-up

Revisions

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Power



LoRa Mesh Node Mini

SolidCraft

Power

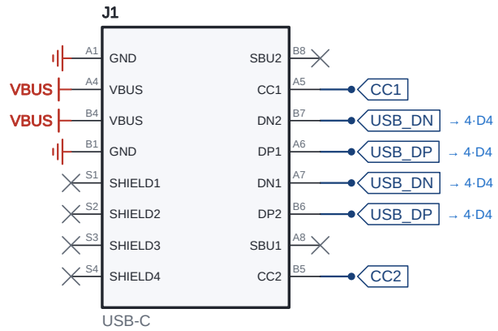
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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SolidCraft example	A3 landscape	phase2.sc

USB





LoRa Mesh Node Mini

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USB

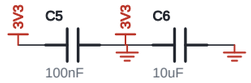
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

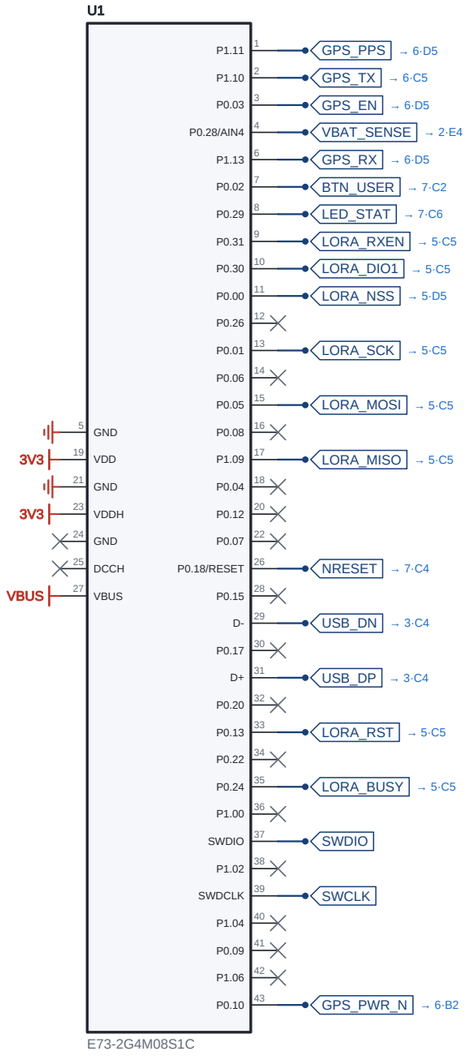
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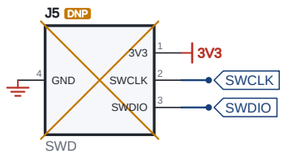
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SolidCraft example	A3 landscape	phase2.sc

☒ DNP = not fitted: pads on the board, no part

MCU







LoRa Mesh Node Mini

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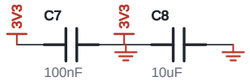
MCU

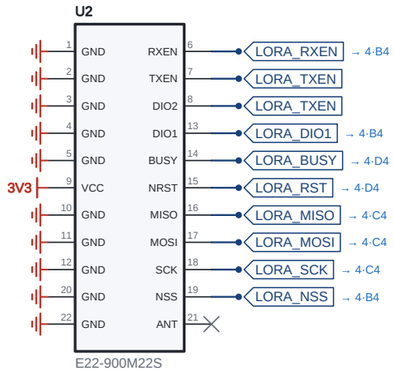
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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LoRa





LoRa Mesh Node Mini

SolidCraft

LoRa

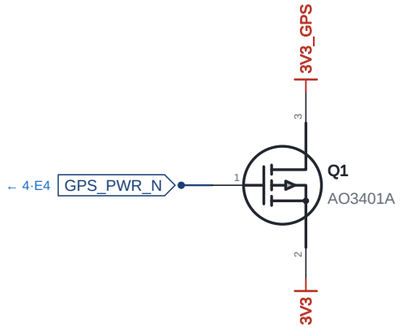
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

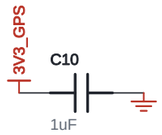
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SolidCraft example	A3 landscape	phase2.sc

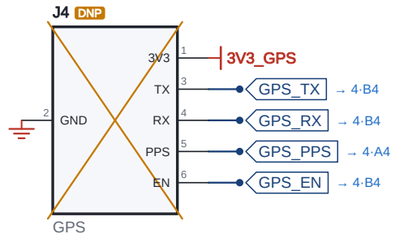
☒ DNP = not fitted: pads on the board, no part

GPS









LoRa Mesh Node Mini

SolidCraft

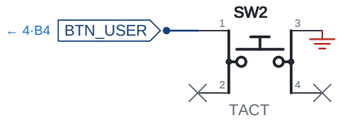
GPS

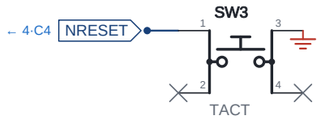
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

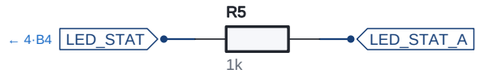
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SolidCraft example	A3 landscape	phase2.sc

UI









LoRa Mesh Node Mini

SolidCraft

UI

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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Signals (bus: shared name prefix, count)

Supply rails: → made in this frame, ← from the page of the frame that makes it; dashed: (detect), (sense) or (pull-up), a signal or reference, not what powers the frame

Power tree (bottom left): feeds detect, sense or pull-up: a signal or reference, not what powers the frame

MCU
U1 E73-2G4M08S1C
J5 SWD-PADS
4 parts · p. 4

3V3 ← p. 2
VBUS (detect) ← p. 3

LoRa
U2 E22-900M22S
3 parts · p. 5

3V3 ← p. 2

GPS
J4 GPS-HDR
Q1 AO3401A
4 parts · p. 6

3V3 ← p. 2

LORA (8)

GPS (5)

VBAT_SENSE

Power
SW1 SLIDE
U3 ME6217C33M5G
U4 TP4054
J2 JST-PH-2
14 parts · p. 2

3V3 → 3 frames
VBUS ← p. 3

USB
J1 USB-C
3 parts · p. 3

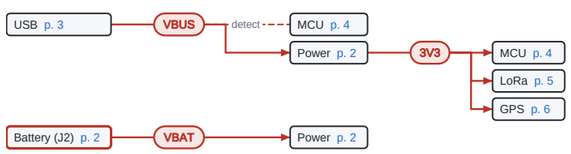
VBUS → p. 2, 4

USB (2)

UI
SW2 TACT
SW3 TACT
4 parts · p. 7

BTN_USER, LED_STAT, NRESET

Power tree



LoRa Mesh Node Mini

SolidCraft

Block diagram

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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SolidCraft example	A3 landscape	phase2.sc

Parts list

30 parts to fit, in 20 lines · 2 not fitted

Qty	Designators	Value	Part	Footprint	MPN · Manufacturer	Description
3	C1–C3	4.7uF	C-4u7	0603	CL10A475KO8NNNC Samsung	
3	C4, C6, C8	10uF	C-10u	0603	CL10A106KP8NNNC Samsung	
3	C5, C7, C9	100nF	C-100n	0603	CC0603KRX7R9BB104 YAGEO	
1	C10	1uF	C-1u	0603	CL10A105KB8NNNC Samsung	
1	D1	RED	LED-RED	LED0603	KT-0603R Hubei KENTO	
1	D2	GREEN	LED-GREEN	LED0603	KT-0603G Hubei KENTO	
1	J1	USB-C	USB-C	USB-C-16P	TYPE-C-31-M-12 Korean Hroparts	
1	J2	Battery	JST-PH-2	JST-PH-2-SMD	S2B-PH-SM4-TB(LF)(SN) JST	
1	Q1	AO3401A	AO3401A	SOT-23	AO3401A AOS	
2	R1, R2	5.1k	R-5k1	0603	0603WAF5101T5E UNI-ROYAL	
1	R3	3.3k	R-3k3	0603	0603WAF3301T5E UNI-ROYAL	
2	R4, R5	1k	R-1k	0603	0603WAF1001T5E UNI-ROYAL	
2	R6, R7	1M	R-1M	0603	0603WAF1004T5E UNI-ROYAL	
1	R8	100k	R-100k	0603	0603WAF1003T5E UNI-ROYAL	
1	SW1	ON/OFF	SLIDE	SLIDE-SPDT	MSK12C02 SHOU HAN	
2	SW2, SW3	TACT	TACT	TACT-5.1	TS-1187A-B-A-B XKB	
1	U1	E73-2G4M08S1C	E73-2G4M08S1C	E73-2G4M08S1C	E73-2G4M08S1C Ebyte	
1	U2	E22-900M22S	E22-900M22S	E22-900M22S	E22-900M22S Ebyte	
1	U3	ME6217C33	ME6217C33M5G	SOT-23-5	ME6217C33M5G Microne	
1	U4	TP4054	TP4054	SOT-23-5	TP4054 UMW	

Continued on the next page

LoRa Mesh Node Mini

SolidCraft

Parts list (1 of 2)

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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SolidCraft example	A3 landscape	phase2.sc

Parts list

30 parts to fit, in 20 lines · 2 not fitted

Qty	Designators	Value	Part	Footprint	MPN · Manufacturer	Description
☒ DNP = not fitted: pads on the board, no part						
1 DNP	J4	GPS	GPS-HDR	HDR-1x6		
1 DNP	J5	SWD	SWD-PADS	SWD-PADS-4		

LoRa Mesh Node Mini — SolidCraft

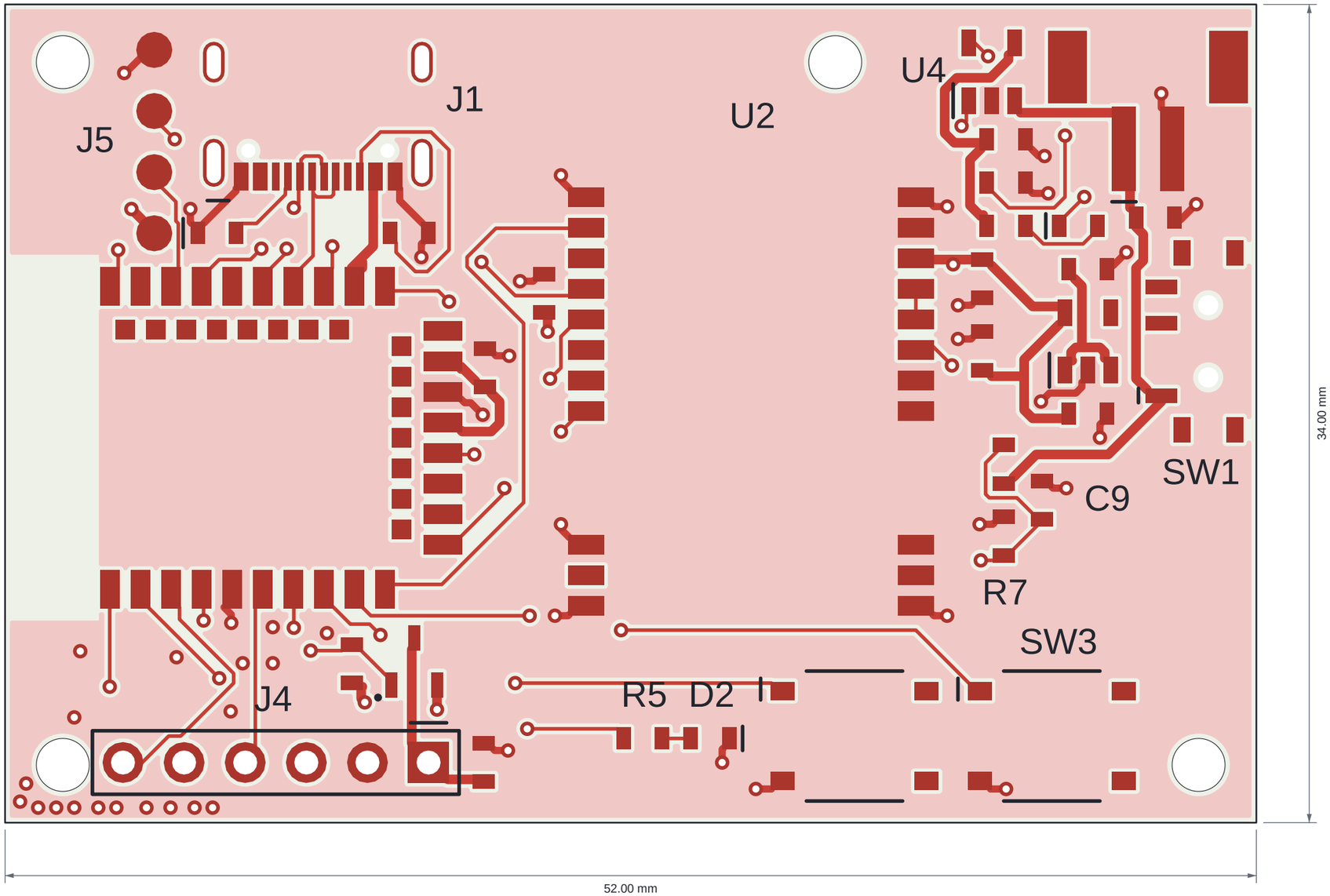
Parts list (2 of 2)

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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SolidCraft example	A3 landscape	phase2.sc

Top copper, pours, pads and vias · top silkscreen · board outline



LoRa Mesh Node Mini

SolidCraft

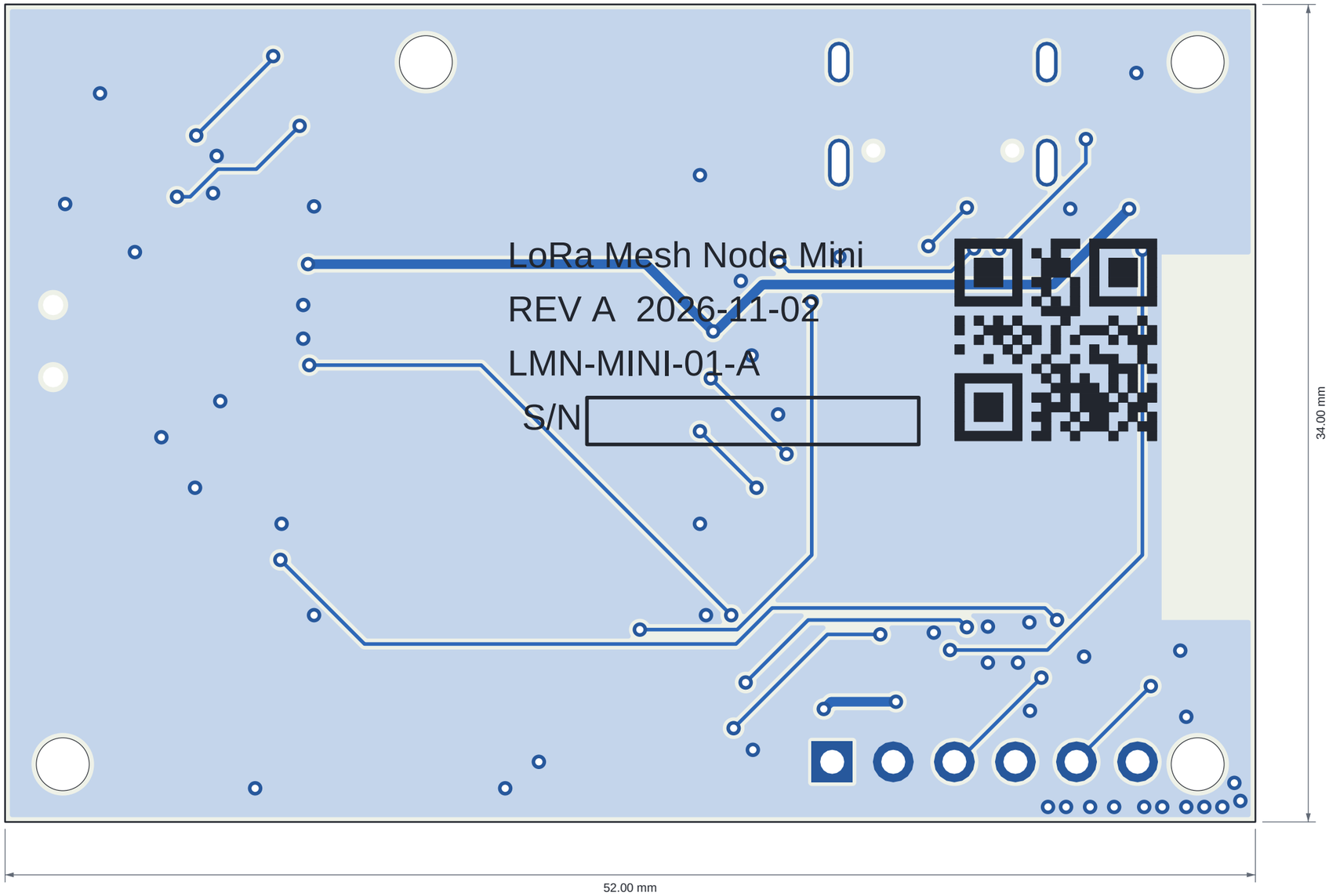
Board top

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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Seen from the bottom (mirrored) · bottom copper · bottom silkscreen



LoRa Mesh Node Mini

SolidCraft

Board bottom (seen from the bottom)

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

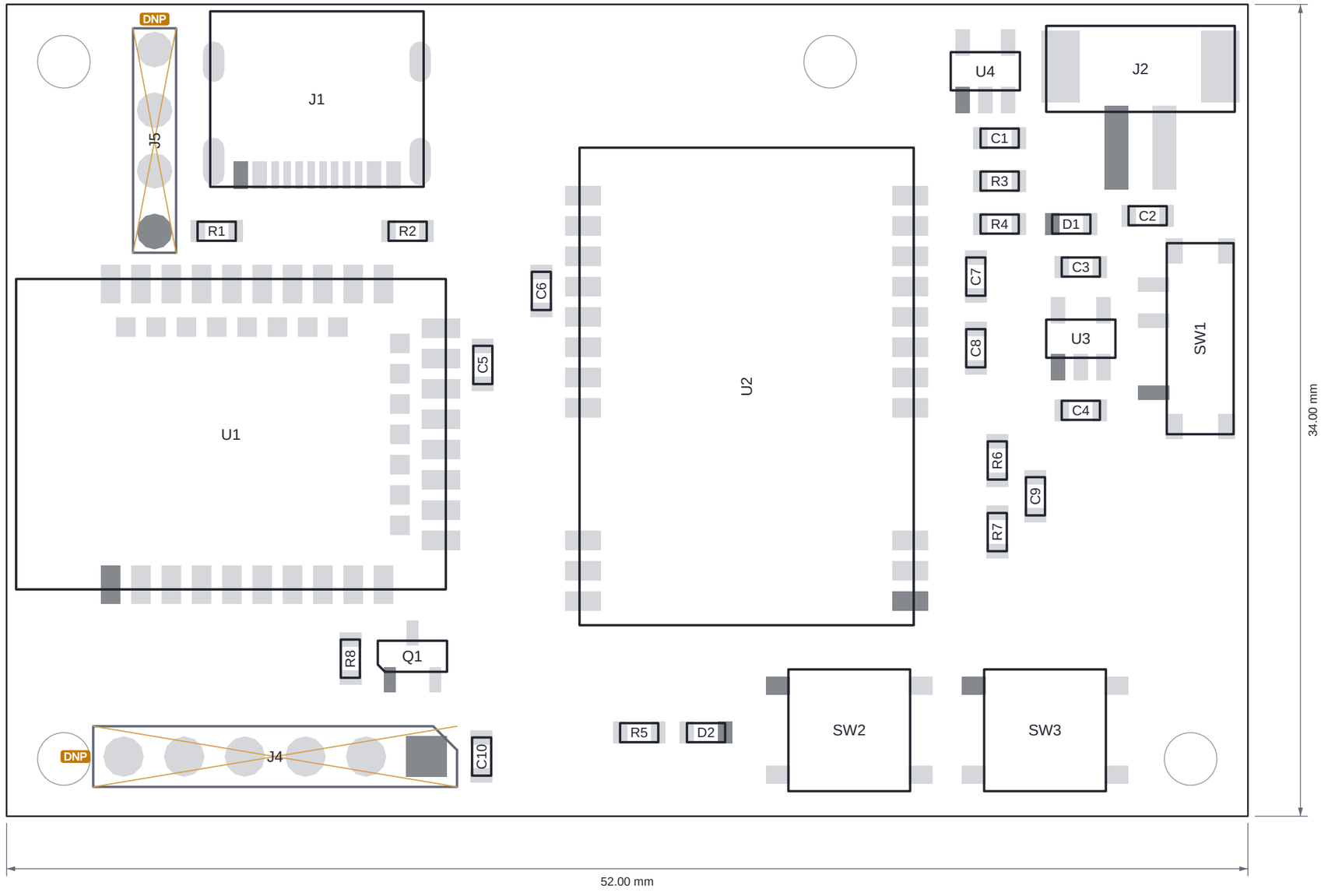
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SolidCraft example	A3 landscape	phase2.sc

Parts on the top: outlines and designators

32 parts on this side (2 not fitted, marked DNP) · pad 1 and cathodes darker · values and part numbers: parts list, page 9

☒ DNP = not fitted: pads on the board, no part



LoRa Mesh Node Mini

SolidCraft

Assembly top

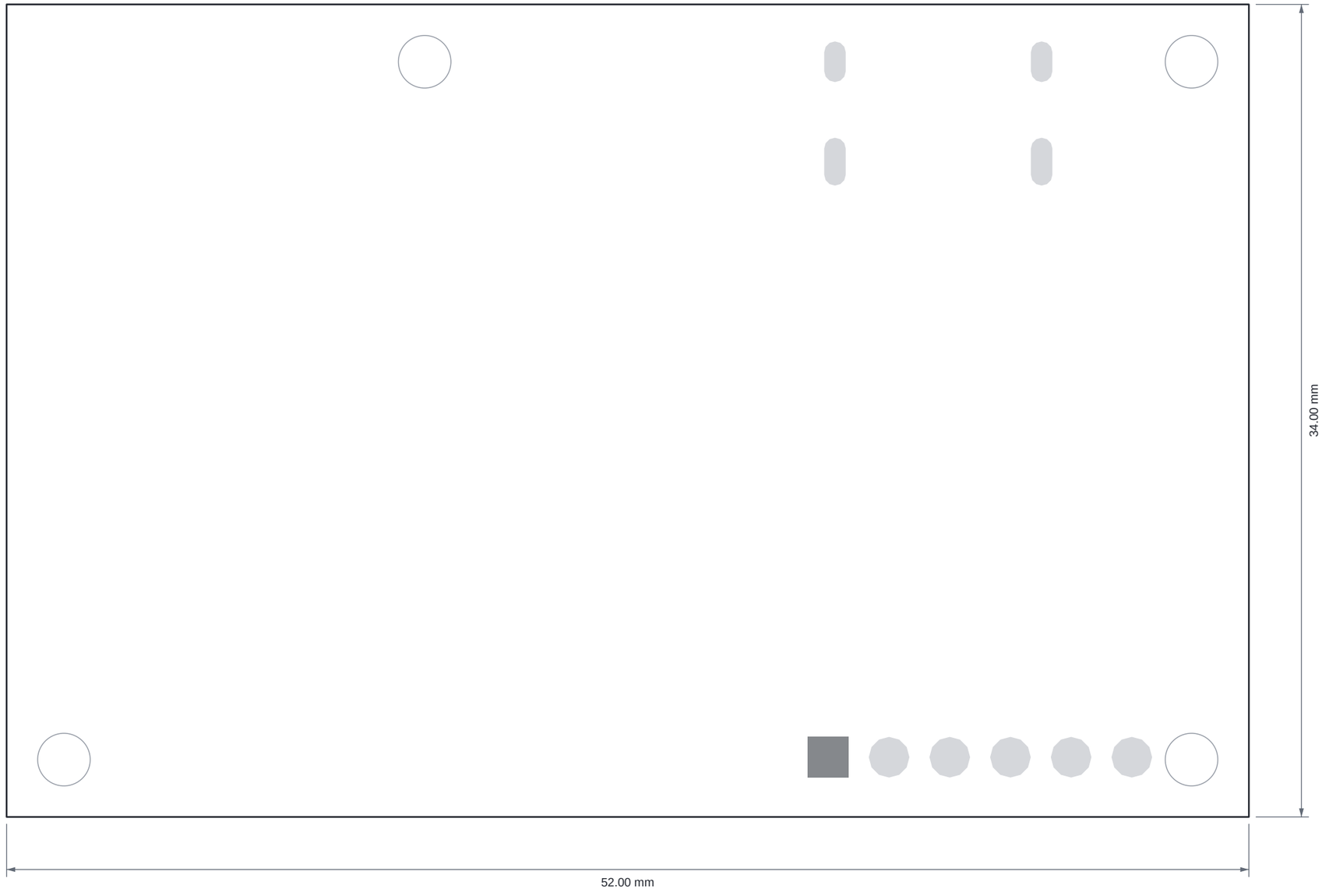
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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Parts on the bottom, seen from the bottom (mirrored): outlines and designators

0 parts on this side · pad 1 and cathodes darker · values and part numbers: parts list, page 9



LoRa Mesh Node Mini

SolidCraft

Assembly bottom (seen from the bottom)

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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Layer stack-up

2 copper layers · 1.600 mm finished (1.60 mm board)

Layer	Kind	Thickness	Copper	εr	tan δ
Top	Copper	0.035 mm	1.0 oz		
Core 1	Core	1.530 mm		4.30	
Bottom	Copper	0.035 mm	1.0 oz		

Net classes on the top layer

Class	Width	Gap	Clearance	Impedance
Power	0.400 mm	–	0.150 mm	118 Ω

Impedance: IPC-2141 microstrip (an edge-coupled pair for a class with one) over the nearest plane. A guide: build to the fab's own figures.

Cross-section (not to scale)



LoRa Mesh Node Mini — SolidCraft

Layer stack-up — nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

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